

AR IN HEALTHCARE

Augmented reality (AR) and virtual reality (VR) are two similar yet different emerging technologies that have gained the attention of many. The gaming and entertainment industries use VR headsets to give users a more immersive experience – something that feels as close to reality as possible.

This article explains how the healthcare industry can use AR/VR to transform the health tech for patients and caregivers.

A Word on Augmented Reality and Virtual Reality (AR/VR)

Many people use the terms AR and VR interchangeably. While there are similarities between the two, they are not one and the same.

Virtual reality uses hardware (headsets) to deliver computer-generated simulations. It is about creating an alternative world and representing it virtually through a headset or other hardware. VR hardware usually comes with sensors that translate real-world actions into a modeled reality.

Augmented reality, on the other hand, joins together real entities with the digital world. You can use AR to represent digital entities or components in the real world. It uses depth tracking and computer vision, and hardware like smart glasses, among other techniques, to display digital content in the real world.

AR/VR Uses in Healthcare

Healthcare systems around the world are beset with several challenges. That is why the healthcare industry has been foremost in embracing new technologies that help overcome challenges related to access, diagnosis, surgery, etc.

One consistent **breakthrough in healthcare sector** is the high demand for services, doctors, and treatment compared to the low supply of healthcare professionals like doctors, nurses, and diagnosticians.

Logistical and financial problems also abound. People who inhabit remote areas like deep mountainous villages also need access to healthcare, but its provision is challenging due to the harsh terrain. More suitable medical equipment is also usually required in such far-flung areas.

Similarly, skilled doctors and medical practitioners are also in short supply, especially in third-world countries. Institutions like **Caribbean medical schools** aim to bridge this gap by training future physicians to serve in underserved regions. As a result, clinics and hospitals often struggle to provide essential health services to patients.

Enhanced Patient Experience

Patient experience is hardly the top concern of healthcare providers when the goal is to save a life. That is why it is often overlooked as being unimportant in the bigger scheme of things. Be that as it may, AR and VR technologies really help improve the patient experience.

Augmented reality solutions can add navigation and pop-up features to enhance the patient experience. AR/VR applications can also use extensions to help patients with chronic pain navigate loneliness during treatment by letting them communicate with their families.

Moreover, patients can use augmented reality to monitor their vitals in a more interactive and engaging way. Patients needing rehabilitation via physiotherapy or other interventions can also use VR headsets to see how the program will roll out.

Psychological Help, Relief & Treatment

Virtual reality solutions can become an exceptionally effective means of treating psychological problems and issues. VR has opened up new opportunities in how therapists deal with patients. For example, you can create intricate virtual environments for smooth and quick healing.

Such virtual environments can help patients improve their pain management skills, streamline sleeping habits, break addictions, and shorten recovery time. In particular, immersive therapies are being piloted to **support opioid addiction treatment**, where

controlled exposure, cue extinction, and real-time biofeedback help patients rehearse responses to triggers in a safe setting. You can use AR/VR to create simulations that can identify disease patterns and help you develop treatment plans that work.

Similarly, VR has emerged as a unique way of dealing with PTSD (post-traumatic stress disorder). It helps treat PTSD patients in two ways: first, you can relax and stabilize the patient via a simulated environment; and two, you can use AR to help your patients learn how to overcome sporadic anxiety episodes associated with PTSD.

Data Visualization & Body Mapping

Access to reliable and consistent patient data is one of the necessary elements for successfully treating a disease or disorder. However, if you don't present patient data in a manner that is easy for doctors and nurses to comprehend, it will hardly deliver any results.

Augmented reality is an excellent technology for visualizing patient information and data. You can show various statistics directly on the patient's body. This is made possible using a pair of smart glasses and a smartphone. The [healthcare web](#) also



Similarly, you can use augmented reality to map a patient's body and use it in case the doctor cannot examine a patient physically or the issue is very complicated.

However, all this would require more technological advancement as it would require you to synchronize and connect various sensory devices and enable hardware configurations.

Improved Diagnostics

Diagnostics is a critical healthcare field upon which the efficacy of treatment plans largely rests. The accuracy and precision of diagnostic tests and results are vital to ensuring the correct diagnosis, which can then be followed up with a suitable treatment plan.

Augmented reality solutions can be of great help in this regard. You can use AR to detect, prevent, and treat several diseases. For example, a doctor can use AR to analyze the patient in layers (veins, organs, bones, lesions, etc.) without penetrating the body. You can also visualize real-time stats from the patient's body in an interactive interface, thus helping the doctor reach a more accurate diagnosis.

HoloLens by Microsoft, an augmented reality/mixed reality device, is an excellent example of how AR can help doctors run simulations and checks using live patient data.

Surgery Assistance

Surgery is another pivotal part of the healthcare industry. AR/VR integration can help surgeons prepare for surgery by providing them with **data visualizations**. Not only does such integration help before the surgery, but AR/VR solutions can also help doctors while performing the surgery to see how it is progressing.



Even the U.S. FDA approves these technologies and vouches for their usefulness in surgical procedures. FDA officials have said that AR/VR capabilities allow medical practitioners to find alternative ways of entering the patient's body, choose relatively less invasive procedures, and streamline workflow. The ensuing benefits include shorter procedures, reduced complications, and improved patient outcomes. However, since AR/VR are emerging technologies, it is still being determined what their full impacts on patients are. For instance, just like motion sickness, a patient wearing a VR headset can suffer cybersickness. Nevertheless, AR/VR still has many benefits to offer to patients.

Moreover, augmented and virtual reality solutions can also be used to train new surgeons. AR/VR can simulate cadavers and replace traditional ones for surgeons to experiment with. They can also simulate live surgery to improve learning outcomes.

In addition, you can use these tools to allow medical students to learn surgical procedures without animal models or cadavers. Not only does AR/VR help doctors, but it also helps patients ready themselves for surgery and thereby reduce their anxiety levels.

Conclusion

Augmented reality and virtual reality come under the umbrella of emerging technologies. These technologies have transformed many industries, from transport and construction to healthcare and finance.

AR/VR applications are improving healthcare systems worldwide. As time goes by and technology advances further, new use cases for AR and VR will emerge. More companies will jump on the AR/VR bandwagon to deliver cutting-edge solutions to their customers.

We have only scratched the surface of how AR and VR are changing the healthcare sector. When combined with other technologies and tools like AI, IoT, and computer vision, the scope is limitless!

Source: [Reference](#)